

Saber Ayatollahi, CFA, MSc

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SUMMARY

CFA charterholder with ~7 years developing and validating quantitative models for multi-asset institutional portfolios - directly relevant to the Senior Manager, Quantitative Strategies and Analytics role in Global Investment Solutions. Holds delegated sign-off authority at Moody's Analytics on valuation and sensitivity outputs for portfolios of \$5-25bn per engagement; performed VaR/CVaR portfolio optimization and risk decomposition, and built stochastic scenario generators, at Ortec Finance. Works daily in Python, SQL, and Excel, automating research and reporting workflows into production pipelines.

CORE SKILLS

Quantitative Investment Analytics (multi-asset) • Model Development, Validation & Assumption Review • Portfolio Optimization - VaR / CVaR • Risk Decomposition & Attribution (Risk Budgeting) • Stochastic & Statistical Modelling, Monte Carlo Simulation • Scenario Analysis & Stress Testing • Python (pandas, NumPy, SciPy) - SQL • R - MATLAB (intermediate, prior research use) - Advanced Excel • Data Analysis & Visualization (Plotly, matplotlib) • Fixed Income & Derivatives Analytics • Process Automation & Analytics Pipelines • Model Governance & Documentation Standards

EXPERIENCE

Moody's Analytics

Asst. Director - Modelling Services

Toronto, May 2023 - Present

Quantitative Model Development & Analytical Tools

- Led design and implementation of an enterprise multi-asset cash flow projection engine supporting base, stress, and reverse-stress scenarios, providing forward-looking analytics that fed asset-allocation and funding-optimization decisions.
- Architected configurable time-bucketed analytics (T+1 through multi-year) enabling funding optimization, refinancing risk assessment, and asset-allocation decisions for institutional investors.
- Embedded behavioral cash flow assumptions, prepayment logic, and macro stress overlays into the projection framework, broadening scenario coverage available to client investment teams.

Model Validation, Assumptions & Performance Assessment

- Holds delegated sign-off authority on valuation, sensitivity, and portfolio-risk outputs for institutional portfolios of \$5-25bn per engagement (~\$50bn cumulative), independently reviewing curve construction, spread calibration, and cross-asset interactions before production release.
- Serves on the model governance committee, contributing to methodology review, benchmarking and documentation standards, and model-performance assessment for client-delivered analytics.
- Assesses model assumptions and underlying data for economic defensibility, escalating outputs lacking mathematical support and acting as escalation point for client-critical analytical issues.
- Validated derivatives pricing outputs across rates, FX, and inflation, and reviewed aggregation logic converting security-level exposures into portfolio-level risk metrics.

Data Analysis, Automation & Stakeholder Reporting

- Re-engineered manual spreadsheet workflows into scalable Python analytics pipelines with embedded logging, validation, and auditability controls.
- Built agentic AI development workflows (Claude Code, Cursor IDE) automating code generation, validation scaffolding, and anomaly detection, cutting development cycle time an estimated 30-40% on comparable modules.
- Prepared analytical summaries for senior stakeholders on rate exposure, scenario impacts, and portfolio-level sensitivities, translating quantitative output into investment-committee-ready narratives.

Client Service Specialist

Toronto, May 2022 - May 2023

- Led onboarding and implementation for U.S. and Canadian pension funds, asset managers, and consulting firms on the PFaroe DB and PFaroe PM analytics platforms, validating model outputs post-deployment.
- Translated investment-team requirements into scoped development requests for product owners; contributed to migrating all assigned client accounts off the legacy Calypso platform.

Ernst & Young (EY)

Senior Consultant - IFRS & Financial Risk Transformation

Toronto, Sep 2021 - Apr 2022

- Delivered IFRS 17 and IFRS 9 transformation projects for Canadian and international insurance clients, aligning measurement requirements with system implementation and financial-reporting frameworks.
- Advised actuarial, IT, finance, and program-management functions and contributed to governance documentation and regulatory-readiness processes.

Ortec Finance Canada

Consultant - Asset-Liability Management & Risk

Toronto, Feb 2019 - Sep 2021

- Performed asset-only and asset-liability (surplus) portfolio optimization on VaR and CVaR using Ortec's GLASS platform, with risk decomposition and near-optimal frontier analysis to test the robustness of allocation recommendations.
- Built and interpreted stochastic economic scenario generators assessing interest-rate risk and funding volatility under base-case and stressed environments.
- Advised institutional clients on Strategic Asset Allocation (SAA) and Tactical Asset Allocation (TAA) trade-offs through stress testing and sensitivity analysis.
- Conducted liability-driven investment (LDI) analysis covering return objectives, duration alignment, and liquidity considerations for pension-fund clients.
- Modelled pension-fund liabilities alongside all major investment-product types and economic variables; presented study findings to client investment committees on-site.
- Led onboarding and the study for a three-plan university pension merger, developing and validating the model capturing all funding-policy dynamics and evaluating duration, currency, inflation, and leverage overlay strategies on a team of three.

EDUCATION & CREDENTIALS

- Chartered Financial Analyst (CFA) - CFA Institute, 2024
- MSc, Financial Modelling - Western University, 2018
- MSc, Chemical Engineering (Full Scholarship) - Western University, 2016